Dominic Parosh Yamarthi

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SUMMARY

AI/ML and Generative AI Engineer with **6+ years of experience** designing and building enterprise AI/ML applications, Python services, cloud-based architectures, and intelligent automation systems. Specialized in **RAG, hybrid retrieval, multi-agent orchestration, semantic search, knowledge graphs, and LLM evaluation**. Developed production GenAI capabilities at Verizon using **LangGraph, LangChain, OpenSearch, FastAPI, and AWS**, contributing to a **40% reduction in test turnaround time**. Experience with **Azure, AWS, Docker, Kubernetes, REST APIs, CI/CD, and distributed data-processing technologies**. Background in financial-document intelligence, predictive analytics, NLP, and evidence-grounded knowledge extraction. Strong ability to translate complex business requirements into scalable AI architectures, evaluate system reliability, and collaborate across engineering and business teams.

EXPERIENCE

Verizon **Apr 2025 – Mar 2026**
Generative AI Engineer – Enterprise AI Platforms *Basking Ridge, NJ*

- **Reduced enterprise test-case turnaround by approximately 40%** by engineering GenTaaS, an AI-powered testing platform integrating LLM generation, retrieval, validation, and enterprise workflow automation using **Python, FastAPI, and LangGraph**.
- Architected **domain-aware RAG pipelines** combining OpenSearch vector retrieval, BM25 keyword matching, embeddings, metadata filtering, and multi-index routing to provide relevant enterprise context for downstream generation tasks.
- Developed **multi-agent orchestration workflows** using LangGraph, implementing specialized generation, critique, validation, and execution stages with conditional routing, tool calling, retries, and human-in-the-loop approval.
- Built and integrated **20+ REST API endpoints** connecting LLM services with Jira, qTest, Tapsium, GitLab, and internal enterprise applications, enabling automated test generation, validation, and downstream execution workflows.
- Developed an **LLM-as-a-judge evaluation framework** comparing generated test cases with reference scenarios through criterion-level scoring, numerical assessments, explanations, and repeated-run consistency analysis.
- Supported AI quality assessment through **golden datasets, SME reviews, correctness checks, semantic similarity evaluation, and prompt regression analysis**; project assessments reported approximately **95% alignment with SME evaluations**.
- Improved application reliability and operational visibility through **structured logging, request tracing, latency monitoring, token-cost tracking, and error analysis**, supporting troubleshooting and optimization of GenAI workflows.
- Contributed to an **AI-powered Order Analyzer and Remediator**, integrating Elasticsearch log retrieval with monitoring, analysis, remediation, and postmortem agent workflows to investigate operational failures and recommend controlled corrective actions.
- Collaborated with QA, product, platform, and engineering teams to translate enterprise requirements into **modular AI services, retrieval architectures, and controlled automation workflows**, supporting containerized delivery and CI/CD practices.

University at Buffalo, SUNY **May 2024 – Mar 2025**
Research Assistant – Artificial Intelligence and Machine Learning *Buffalo, NY*

- Developed reproducible **Python, PyTorch, and TensorFlow** research pipelines for multimodal deceptive-review detection, integrating Transformer-based text representations, visual features, and supervised classification.
- Investigated **knowledge representation, entity-relation extraction, evidence verification, and knowledge-graph approaches** for grounding model outputs in identifiable source information.
- Built data preprocessing, feature engineering, model comparison, and evaluation workflows for multimodal learning and transportation-related predictive analytics.
- Conducted experimental analysis of model correctness, classification errors, feature quality, and evidence consistency to improve research reproducibility and understand prediction failures.

University at Buffalo, SUNY **Jan 2024 – May 2024**
Graduate Teaching Assistant – Data Intensive Computing *Buffalo, NY*

- Supported **200+ students** in Python programming, large-scale data processing, distributed-computing concepts,

debugging, and software development practices.

TechNVision / Emagia **Aug 2018 – Dec 2022**
Machine Learning Engineer – Financial Technology *Hyderabad, India*

- Developed enterprise **intelligent document processing pipelines** for financial applications using Python, OCR, NLP, classification, named-entity recognition, and structured information extraction.
- Built data ingestion, preprocessing, and validation workflows transforming noisy invoices and financial documents into structured records for downstream enterprise processing and analytics.
- Integrated machine learning inference services with enterprise applications through **Python REST APIs**, supporting modular service architectures and Azure-hosted deployments.
- Contributed to document-processing solutions supporting **500+ users across 90+ countries**, improving the automation and accessibility of financial-document information.
- Collaborated with software engineers and business stakeholders to investigate model errors, improve extraction quality, and maintain reliable integration with financial workflows.

SKILLS

Programming & Backend: Python, SQL, FastAPI, Flask, REST APIs, Microservices, Object-Oriented Programming, API Integration, Git, Software Architecture, Unit Testing, CI/CD

Generative AI & LLMs: Large Language Models, LangChain, LangGraph, RAG, Agentic RAG, Prompt Engineering, Embeddings, Semantic Search, Structured Outputs, Tool Calling, Multi-Agent Systems, Human-in-the-Loop Workflows

Knowledge Graphs & Retrieval: Knowledge Graphs, Neo4j, GraphRAG Architecture Patterns, Entity-Relation Extraction, Semantic Modeling, OpenSearch, Elasticsearch, FAISS, Pinecone, ChromaDB, Hybrid Search, BM25, Vector Similarity, Multi-Index Routing

Machine Learning & Analytics: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NLP, Predictive Modeling, Regression, Classification, Anomaly Detection, Feature Engineering, Statistical Analysis, Model Evaluation

Cloud & Distributed Systems: AWS, Amazon Bedrock, SageMaker, S3, SQS, Azure OpenAI, Azure AI Search, Docker, Kubernetes, Linux, Cloud-Native Applications, Kafka, Event-Driven Architecture Concepts

Data Engineering & Databases: ETL/ELT, Pandas, NumPy, PySpark, Apache Spark, SQL, PostgreSQL, MySQL, MongoDB, Redis, Data Validation, Data Transformation, Structured and Unstructured Data Processing

AI Evaluation & Observability: LLM-as-a-Judge, Golden Datasets, Semantic Similarity, SME Alignment, Retrieval Evaluation, Prompt Regression Testing, Quality Metrics, Structured Logging, Latency Monitoring, Token-Cost Monitoring, Guardrails

PROJECTS

Evidence-Grounded Knowledge Graph Extraction – Python, Neo4j, LLMs

- Designed a research-oriented **knowledge extraction workflow** identifying entities, relationships, and supporting evidence from unstructured documents, exploring graph-based representations for semantic reasoning.
- Investigated **evidence verification, provenance tracking, and abstention mechanisms** to reduce unsupported entity relationships and improve the reliability of extracted knowledge.
- Explored **GraphRAG-oriented retrieval patterns** combining entity relationships with source-document evidence to support more explainable, context-grounded responses.

GenAI Evaluation and Quality Measurement – Python, LLM-as-a-Judge

- Built an offline evaluation workflow comparing LLM-generated test cases against reference cases, returning **criterion-level quality scores, overall numerical assessments, and explanatory rationales**.
- Evaluated consistency across repeated executions and investigated differences in generated outputs to support repeatable quality measurement and model improvement.

Multimodal Fake Review Detection – PyTorch, Transformers, NLP

- Developed multimodal classification experiments combining **Transformer-based textual representations and visual features** to identify deceptive review patterns.
- Prepared real and synthetic review datasets, evaluated classification models, and analyzed representation quality and prediction failures.

Recommendation System MLOps – Python, Machine Learning

- Built reproducible data preprocessing, model training, and evaluation workflows for recommendation systems, using **Precision@10, Recall@10, NDCG, RMSE, and MAE** to measure predictive and ranking quality.

United States Road Accident Prediction – Python, Random Forest

- Developed a predictive analytics workflow using **Random Forest regression**, traffic-related data, feature engineering, and model evaluation to investigate roadway accident patterns and contributing factors.

EDUCATION

University at Buffalo, State University of New York – M.S. in Data Science 2024

Gayatri Vidya Parishad College of Engineering (Autonomous) – B.Tech. in Computer Science 2018

PUBLICATIONS

United States Road Accident Prediction using Random Forest Predictor. arXiv:2505.06246 (preprint, 2025).